

Orexivive: Turning Fatigue into Flourish through Orexin-1 Activation

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The disease : Atypical Depression

Key Clinical Features:

- Mood improves temporarily with positive events
- Hyperphagia & weight gain
- Hypersomnia
- Leaden paralysis in arms and legs
- High interpersonal rejection sensitivity
- Commonly co-occurs with anxiety; risk increased by female sex, smoking, social isolation, sedentary lifestyle, chronic illness

Treatment gap: no drugs directly target hypersomnia, weight gain, or rejection sensitivity.

Market Considerations:

Economic burden - Depression cost the U.S. \$236 billion in 2018

Indirect costs (ex: workplace productivity losses) exceed medical costs

Early onset & chronicity of atypical depression further increase burden

Antidepressant market growth - Valued at \$18.7B in 2024, projected to reach \$37.9B by 2034 (CAGR ~7.5%)

Demand rising for treatments addressing treatment-resistant & underserved subtypes

Antidepressant Drugs Market Size & Share, Forecast Report 2021-2027. (2021, September). Global Market Insights Inc.
<https://www.gminsights.com/industry-analysis/antidepressant-drugs-market>

Singh, T., & Williams, K. (2006). Atypical Depression. *Psychiatry (Edgmont)*, 3(4), 33. <https://pmc.ncbi.nlm.nih.gov/articles/PMC2990566/>

Wandler, K. (2024, September 17). *SSRIs vs MAOIs: How These Antidepressants Compare* | *The Recovery Village Cherry Hill at Cooper Drug and Alcohol Rehab*. The Recovery Village Cherry Hill at Cooper.
<https://www.southjerseyrecovery.com/resources/ssris-vs-maois/>

Hypothetical Mechanism of Action

Orexive crosses blood-brain barrier

Selectively binds
Orexin-1 receptors -
OX1R (not OX2R)



Calibrated OX1R stimulation steadies the fight-or-flight system—supporting stress resilience—Orexive is partial agonist, thus it avoids anxiety and hyper-vigilance

The net effect is increased wakefulness and attention

Orexin-A activates OX1R on locus coeruleus neurons, depolarizing them to increase firing and thereby boosting noradrenaline release to the cortex.

These effects energize people, motivating them to engage more socially – it reduces mood swings and sensitivity to rejection, thereby alleviating atypical depression.

Decreases hypersomnia by reducing REM (Rapid Eye Movement) sleep



Action

OX1R has higher affinity for orexin-A (peptide made in the hypothalamus - binds to orexin receptors)

Activation of OX1R by orexin-A depolarizes cholinergic neurons in the pedunculo pontine and laterodorsal tegmental nuclei, thereby boosting Acetylcholine release to the brain. This effect is blocked by the selective OX1R antagonist SB-334867 (test tool - helps prove whether OX1R is really responsible for an effect)

Drug Discovery

- Orexivive = Orexin (target) + Vive (Latin: life/energy)
- Focus groups linked the name with vitality, wakefulness, motivation
- Rated highly for being easy to pronounce, memorable, uplifting

- Identified through computational drug design – AI screened thousands of molecules for OX1R binding
- Lead compound chosen for high selectivity to OX1R and low activity on OX2R
- Progressed from in vitro / in vivo success → formulation development

Step 1: Core scaffold from vanillin – Vanillin, a renewable compound and a lignin derivative, is used as the chemical backbone.

Step 2: Functionalization: The vanillin scaffold is modified using catalytic hydrogenation and amide coupling.

Step 3: Purification – The final compound is isolated using high-performance liquid chromatography (HPLC) to achieve >99% purity.

The final drug is formulated into a once-daily oral capsule with controlled release technology, ensuring stable plasma levels over 24 hours.

Pharmaceutical development

Route of delivery

- Oral first line (once-daily a.m. dose).
- Back-up options: ODT (orally disintegrating tablet) for patients with morning inertia; intranasal micro-spray for rapid “wake-initiation” on bad days.



Hypothetical pharmacological features

- Partial agonist ($E_{max} \approx 60\%$) = it turns the OX₁R “on,” but only to ~60% of full power
- Onset 30–60 min = patients should feel effects within an hour.
- $t_{1/2} \sim 20$ h = the half-life is ~20 hours, so one dose keeps levels steady through the day
- Safety profile (expected): mild, transient insomnia or headache at high dose

Dosing frequency

Start 25 mg qAM $\times 7$ days $\rightarrow$ 50 mg qAM; consider 100 mg after 2–3 weeks if symptoms persist

Once daily maintains daytime energy and reduces naps; avoid evening dosing to prevent insomnia.

Formulation / dosage form

XR (extended release) capsule with pH-independent multiparticulate beads (contains many tiny coated beads that release drug slowly) $\rightarrow$ flat PK across meals (steady concentration in the blood through the day)

Strengths: 25 mg, 50 mg, 100 mg; color-coded to avoid mix-ups.

Current clinical development progress

Phase I — First-in-human (healthy adults)

Design: Randomized, double-blind, placebo-controlled; SAD (5–200 mg) + MAD (25, 50, 100 mg qAM ×14 days), n=96.

PK: Rapid absorption (T_{max} ~1 h), $t_{1/2}$ ~20 h, dose-proportional exposure; minimal food effect (XR beads).

CNS effects: Increased alertness on psychomotor vigilance task at ≥50 mg without jitteriness; no next-day impairment.

Safety/Tolerability: Mostly mild headache, dry mouth, transient ↑Heart Rate (3–5 bpm); no clinically relevant BP or QTc signals; no euphoria/likability on abuse-liability scales.

Efficacy: On a standard depression scale (IDS-SR), the drug group improved 5.4 points more than placebo after 8 weeks—considered clinically meaningful.

Atypical symptom cluster: Hypersomnia down: People were less sleepy in the day (Epworth score dropped by 4.1)

Daytime naps down 38%.

Rejection sensitivity down (RSQ fell by 22%)

CGI-I responders 62% vs 34% on placebo

Weight-neutral overall (drug group -0.3 kg vs placebo +0.8 kg).

People also reported fewer evening cravings

Side effects were similar to Phase 1 and mostly mild: headache (11%), dry mouth (8%), and mild insomnia (6%) mainly in the first weeks.

Phase IIa — Proof of concept in atypical depression

Design: 8-week, multi-site, double-blind Orexive 50→100 mg qAM vs placebo; n=132 adults (Major Depressive Disorder with atypical features)

Primary endpoint: Change in IDS-SR (Inventory of Depressive Symptomatology – Self-Report (a standard questionnaire patients fill out) total with atypical subscale.

Key secondaries: Epworth Sleepiness Scale, Rejection Sensitivity Questionnaire, clinician-rated CGI-I (Clinician's Global Impression–Improvement).

Plan for continuation of clinical development

Phase IIb — Dose-finding & confirmatory signal (atypical MDD)

Objective: Confirm efficacy on atypical cluster (hypersomnia, rejection sensitivity) and select registrational dose.

Design: 12-week, multi-country, double-blind, randomized placebo-controlled.

Arms (qAM): Placebo • Orexive 50 mg • Orexive 100 mg.

N ≈ 360 adults (age: 18–65) with MDD w/ atypical features; hypersomnia ≥10 h/night or ≥3 naps/wk required.

Key endpoints: Primary: IDS-SR total (hierarchical test includes atypical subscale).

Key secondary: Epworth Sleepiness Scale, Rejection Sensitivity Questionnaire, CGI-I responders.

Phase III — Registrational program

Trial A: Global efficacy & safety

Design: 12-week, double-blind, active-controlled and placebo-controlled; N ≈ 800.

Arms: Placebo • Orexive [selected dose] • SSRI comparator (sertraline) to benchmark effect.

Primary: IDS-SR total at Week 8 (confirmatory) with Week 12 maintenance.

Key secondaries: Atypical subscale, SDS functioning, CGI-I, time to first daytime nap.
Stats: Mixed-model repeated measures; dose-response via MCP-Mod

Trial B: Relapse-prevention & functioning

Design: 28-week; 8-week open-label lead-in → randomized withdrawal (Orexive vs placebo) for 20 weeks.

Primary: Time to relapse (predefined IDS-SR + clinician confirmation).

Key secondaries: Work Productivity and Activity Impairment (WPAI), social participation index.

Conclusion

Orexivive is a selective OX₁R partial agonist designed to target the atypical symptom cluster—hypersomnia, rejection sensitivity—while avoiding sedative trade-offs.

Early clinical signals show reduced daytime sleepiness/naps, better mood reactivity & social engagement, and no major safety concerns.

The mechanism (wakefulness + motivation, moderated stress) directly maps to unmet needs in atypical MDD.

If successful, Orexivive could become the first treatment purpose-built for this under-served subtype.

SMILING-DEPRESSION

The mask may hide suffering



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We're seeking support to complete Phase IIb dose-finding and Phase III registrational trials, plus CMC scale-up and long-term safety.

With your backing, we can advance a novel, targeted therapy toward patients who need it most.